THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET A

Chapter 1: **Exploring the Investigative World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a systematic investigation, a scientist begins with: 1

(a) a final conclusion **(b)** a focused question
(c) an exam paper **(d)** a printed result

2. Which of these can be changed to study puri puffing? 1

(a) temperature of the oil **(b)** size of the kitchen
(c) name of the cook **(d)** colour of the plate

3. Keeping notes of everything seen, smelt and sensed during an experiment helps in: 1

(a) skipping measurements **(b)** changing the results
(c) avoiding the experiment **(d)** making sense of the results later

4. Which question is a scientific question about puris? 1

(a) Is puri better than rice?
(b) Does the thickness of dough change how the puri puffs?
(c) Who likes puris most?
(d) Which puri is the prettiest?

5. In Grade 6 the book began with: 1

(a) only formulas **(b)** only experiments
(c) wonder through why and how questions **(d)** only history

6. The kite on right-hand pages symbolises: 1

(a) a festival only **(b)** a map
(c) a weapon **(d)** curiosity taking flight

7. Ecosystems are formed by living things interacting with their: 1

(a) only books **(b)** only machines
(c) environment **(d)** only each other's names

8. Mirrors and lenses are studied under the topic of: 1

(a) light **(b)** sound **(c)** heat only **(d)** magnets only

9. A measurement should always be written with its: 1

(a) shadow **(b)** unit **(c)** colour **(d)** nickname

10. If you prick a hole in dough before frying, you are testing a: 1

(a) safe practice rule **(b)** conclusion
(c) measuring unit **(d)** new variable

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is a variable in an experiment? Give one example from puri frying. 2

12. State why only one thing should be changed at a time. 2

13. Write two things that can be controlled when making a puri. 2

14. Write one yes/no and one numerical observation for a puri experiment. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A student wants to know whether oil temperature affects puffing. 3
(i) What will she change?
(ii) What will she measure?
(iii) What should she keep the same?

16. Explain the difference between things we can control and things we observe in an experiment, with one example of each. 3

17. List three factors that may affect how a puri puffs. 3

OR

Why is it wrong to change both oil temperature and flour type in the same trial? 3

18. Describe the steps of a simple investigation, starting from a question. 3

OR

How does the chapter link the Moon's phases to calendars? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Design an investigation to find how oil temperature affects puffing of puris. Write 5
(i) the question,
(ii) the variable,
(iii) the fixed factors,
(iv) the measurements and
(v) how you will record results.

OR

Explain with an example what a fair test is. Why does a fair test need controlled conditions, and what can go wrong if it does not? 5

20. Describe the process of systematic investigation in five steps using a kitchen example. 5

OR

Explain how observing the Moon helped create calendars and how this shows science links Earth and sky. 5

END OF QUESTION PAPER — Total: 40 Marks